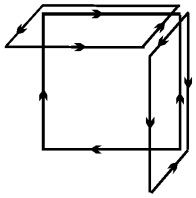
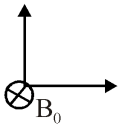
SOLUTION
SECTION-I

Single Correct Answer Type **7 Q. [3 M (–1)]**

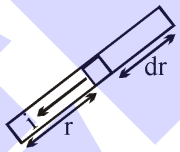
1. Ans. (D)

Sol. Imagine 3 loops.  

$B = \sqrt{B_0^2 + B_0^2 + B_0^2} = \sqrt{3}\,B_0$ towards F

2. Ans. (A)

Sol. Torque due to magnetic force $|d\vec{\tau}| = \int_0^L (idrB)r = \frac{iL^2B}{2}$

In equilibrium $\frac{iL^2B}{2} = (kx) \times L\sin 30° \text{ or } x = \frac{5iLB}{8k}$ 

3. Ans. (C)

Sol. Along y $v_y^2 = 2 \times \frac{qE}{m} \times y$

Net speed = $\sqrt{v^2 + v_y^2}$

4. Ans. (A)

Sol. $B_E - B_P = B_1$ $B_E + B_P = B_2$

$B_E = \frac{B_1 + B_2}{2}$

$B_P = \frac{B_2 - B_1}{2} = \frac{\mu_0 J}{2}$

So answer is = $J\left(\frac{B_1 + B_2}{2}\right)$

5. Ans. (D)

Sol. Current sensitivity = $\frac{NAB}{C}$

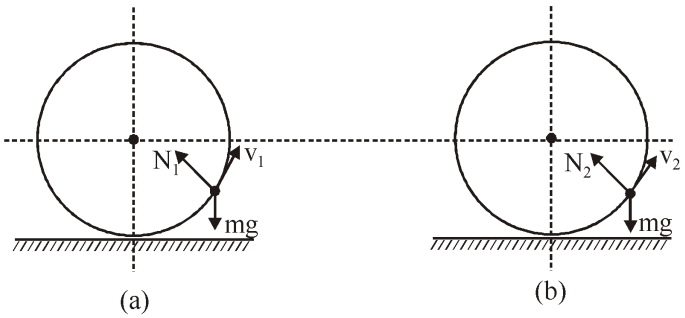
6. Ans. (B)

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7. Ans. (A)

Sol.



(a) (b)

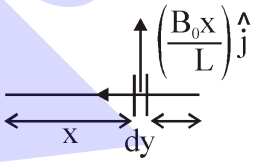
F.B.D in two cases is as shown in case (b) normal will be less and incase (b) work done by magnetic field is zero so height in two cases will be same.]

Multiple Correct Answer Type

9 Q. [4 M (–1)]

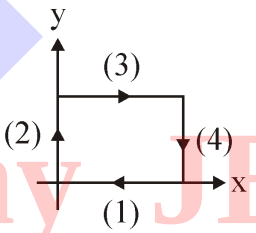
8. Ans. (ABCD)

Sol. Wire segment (1)



$$(d\vec{f}) = \int_0^L i\,dx\left(\frac{B_0x}{2}\right) = \frac{iB_0L}{2};\ d\vec{f} = \frac{-iB_0L}{2}\hat{k}$$

Similary for $\vec{f}_2 = -\frac{-iB_0L}{2}\hat{k};\ \vec{f}_3 = \frac{iB_0L}{2}\hat{k};\ \vec{f}_4 = \frac{iB_0L}{2}\hat{k}\ \ \vec{f}_n = 0$



If coil is constrained to rotate about y-axis $|\vec{\tau}| = i\left(\frac{B_0L}{2}\right)L = \frac{iAB_0}{2}$

$\Rightarrow$ Similarly for torque about torque has same magnitude

9. Ans. (C,D)

Sol. Current = $\frac{(4q)\omega}{2\pi}$ for each face from which axis is passing

$$B = 2\times\frac{\mu_0 iR^2}{2\left(R^2+x^2\right)^{\frac{3}{2}}} = \frac{2\times\mu_0(4q)\omega\left(\frac{a}{\sqrt{2}}\right)^2}{2\cdot2\pi\left[\left(\frac{a}{\sqrt{2}}\right)^2+\left(\frac{a}{2}\right)^2\right]^{\frac{3}{2}}} = \frac{8\mu_0q\omega}{3\sqrt{3}\pi a}$$

Reversing any of the four charges mass the current either zero or in opposite sense at two surface.

10. Ans. (A, D)

Sol. (A) $\frac{\mu_0K}{2}$ hence $K = \sigma V$

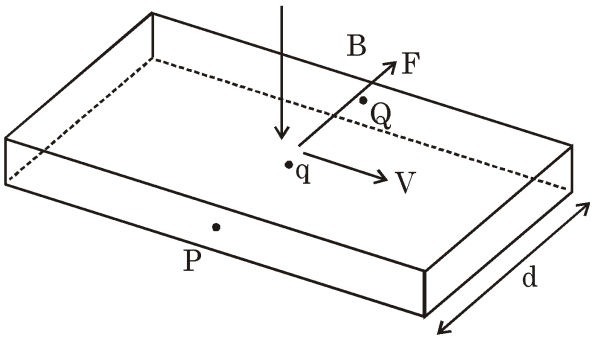
(D) $F = q\vec{u}\times\vec{B}$ hence upwards.

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11. Ans. (A, B)

Sol.



$\therefore$ Charges separate between P & Q

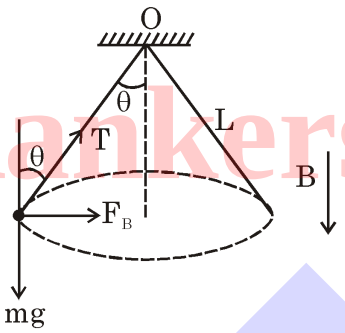
hence potential difference develops between P & Q.

$\Delta V = Ed = v_d Bd$

$\therefore$ if 'B' increases, ΔV increases.

12. Ans. (A,D)

Sol.



$T\cos\theta = mg$ ... (i)

$T\sin\theta + F_B = mw^2 (L\sin\theta)$ .... (ii)

From equation (i) and (ii)

$$\left(\frac{mg}{\cos\theta}\right)\sin\theta + q(\omega L\sin\theta)B$$

$$= m\omega^2 L\sin\theta$$

$$\Rightarrow \quad B = \frac{m\omega}{q} - \frac{mg\sin\theta}{\cos\theta\cdot(\omega L\sin\theta)}$$

$$B = \frac{1}{\beta}\left[\omega - \frac{g}{\omega L\cos\theta}\right], \left(\because \beta = \frac{q}{m}\right)$$

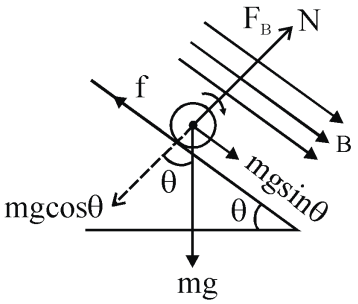
Since torque due to force mg and F_B is not zero about 'O'. Hence angular momentum is not conserved about point of suspension 'O'.

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13. Ans. (B,C,D)

Sol.



Using $\vec{F}_B = i\left(d\vec{\ell} \times \vec{B}\right)$, we can find direction of F_B, as shown in figure.

for angular velocity 'ω' to be constant,

$\tau_{cm} = 0$

$\Rightarrow$ $f = 0$

$\Rightarrow$ $N = 0$

$\Rightarrow$ $F_B = mg\cos\theta$

Due to component of mg, i.e., mgsinθ

translational velocity will change.

14. Ans. (A,D)

15. Ans. (A, C, D)

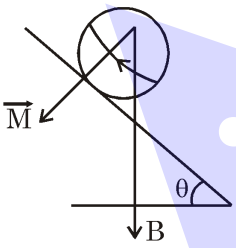
Sol. $I\vec{\alpha} = \vec{\tau}$

$\vec{\tau} = \vec{\mu} \times \vec{B}$

$\vec{\mu} = \pi R^2 \times I\left(-\hat{k}\right)$ & $I = \dfrac{MR^2}{2}$

16. Ans. (B,C)

Sol.



$MB\sin\theta = mg\sin\theta \times R$

$B = \dfrac{mg}{\pi iR}$

Linked Comprehension Type (1 Para × 3Q.) (1 Para × 2Q.) [3 M (-1)]
(Single Correct Answer Type)

17. Ans. (D)

Sol. Applying Gauss law, $\dfrac{-ne}{\epsilon_0}\pi r^2\ell = E2\pi rl \Rightarrow \vec{E} = \dfrac{-ner}{2\epsilon_0}\hat{r}$

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18. Ans. (B)

Sol. Applying Ampere's law : $\oint \vec{B}.d\vec{l} = \mu_0 I$, $B2\pi r = \mu_0 ne\pi r^2 v \Rightarrow B = \frac{\mu_0 ner}{2} v$

19. Ans. (B)

Sol. Net force on electron = $q\vec{E} + q\vec{v} \times \vec{B} = \frac{ne^2 r}{2\,\epsilon_0}\left[1 - \mu_0 \varepsilon_0 v^2\right]\hat{r}$

20. Ans. (D)

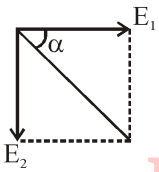
Sol. $qv_dB = qE$

$$E = v_d B = \left(\frac{i}{nqA}\right)B = \frac{iB}{nq(YZ)}$$

$$\Delta V_H = E \cdot Y = \frac{iB}{nq(Z)}$$

21. Ans. (A)

Sol.



$$\tan\alpha = \frac{E_2}{E_1} = \frac{E_2}{\left(\frac{J}{\sigma}\right)} = \frac{\left(\frac{iB}{nqA}\right)}{\frac{J}{\sigma}} = \frac{B\sigma}{nq}$$

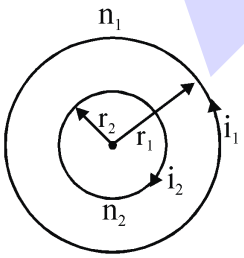
SECTION-II

Numerical Answer Type Question **2Q.[3M (0)]**

(upto second decimal place)

22. Ans. 216.00

Sol. Energy stored permit length = $\frac{B^2}{2\mu_0} \times \pi\left(r_1^2 - r_2^2\right)$



23. Ans.160.00

Sol. $r = \frac{mv}{qB} = \frac{\sqrt{2mK}}{eB}$

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$$0.75 = \frac{\sqrt{2\times16\times10^{-27}\times8\times10^{6}\times1.6\times10^{-19}}}{1.6\times10^{-19}\times B}$$

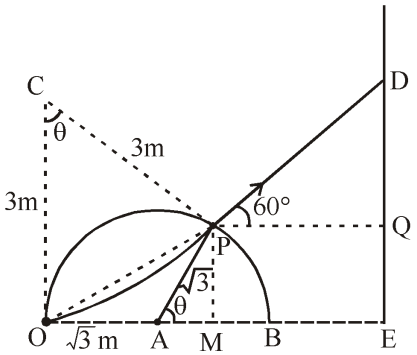
$$\Rightarrow B = \frac{4}{7.5} \text{ so } 300B = 160 \text{ T}$$

SECTION-III

Numerical Grid Type (Ranging from 0 to 9) **4 Q. [4 M (0)]**

24. Ans. 3

Sol.



Considering the upper half of the area

Radius of paticle is circular motion $R = \frac{mv}{qB} = \frac{(2\times10^{-3})(0.3)}{(1\times10^{-3})\times0.2} = 3$ m

Radius of circular arc, $r = \sqrt{3}$ m

from ΔCOP tand ΔOAP

$$OP = 2\times3\times\sin\frac{\theta}{2} = 2\times\sqrt{3}\sin\left(\frac{180-\theta}{2}\right)$$

$$\Rightarrow \tan\left(\frac{\theta}{2}\right) = \frac{1}{\sqrt{3}} \Rightarrow \theta = 60°$$

From, ΔAPM,

$$PM = \sqrt{3}\sin 60° = \frac{3}{2} = \theta E$$

$$AM = \sqrt{3}\cos 60° = \frac{\sqrt{3}}{2}$$

From diagram,

$$PQ = ME = \sqrt{3} + \frac{\sqrt{3}}{2} = \frac{3\sqrt{3}}{2}$$

From ΔPQD,

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$$\tan 60° = \frac{DQ}{PQ} \Rightarrow DQ = \sqrt{3} \times \frac{3\sqrt{3}}{2} = \frac{9}{2} m$$

$$\therefore DE = QE + DQ = \frac{3}{2} + \frac{9}{2} = 6m$$

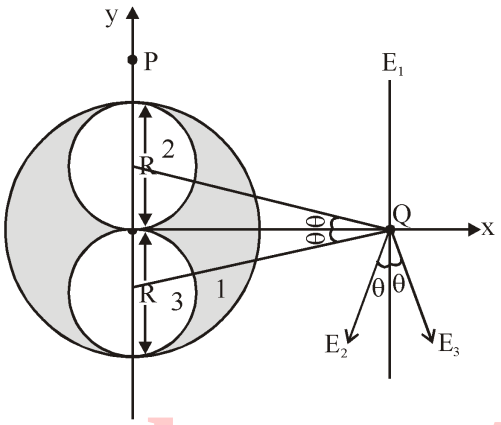
Similarlly we can get, from lower half

$\therefore$ Distance between the points where charges will strike after deflection = 6 + 6 = 12m = 4x

$\Rightarrow x = 3$

25. Ans. 9

Sol.



$$\tan\theta = \frac{R}{2R} = \frac{1}{2}$$

Assume hollow cylinders as current density ± J and find E_1,E_2 & E_3 individually finally add all of them

26. Ans. 5

Sol. $\oint B \cdot d\vec{\ell} = \mu_0 i_{net}$

$$B \times 2\pi r = \mu_0 \int_a^r \left(J_0 \frac{r}{a} \right) 2\pi r^2 dr$$

27. Ans. 1

Sol : Consider unit length of cylinder charge = $\sigma 2\pi R \times 1$

time = $2\pi/\omega$

$$\text{So} \quad ni = \frac{2\pi R\sigma\omega}{2\pi} = \sigma R\omega$$

Now $B = \mu_0 ni$

$B = \mu_0 \sigma R\omega$

SECTION-IV

Matrix Match Type (4 × 5) **1 Q. [8 M (for each entry +2(0)]**

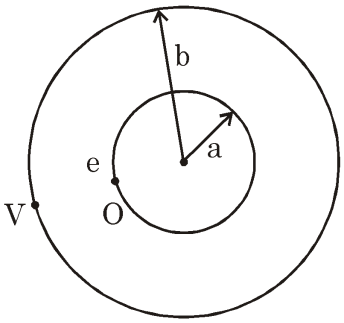
28. Ans. (A) P,S; (B) Q,T,R; (C) Q,T,R; (D) P

Subjective Type **2 Q. [4 M (0)]**

29. Ans. (i) $u = \sqrt{\frac{2eV_0}{m_e}}$, (ii) $\frac{2b m_e u_0}{e(b^2 - a^2)}$

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Sol.

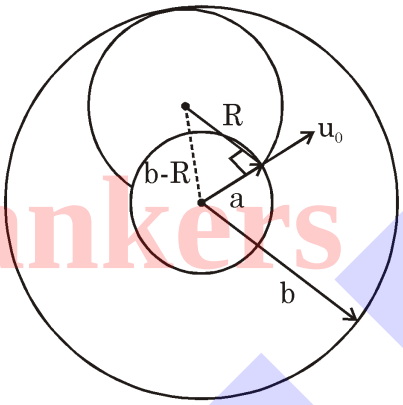


(i) From work-energy theorem between inner and outer cylinder.

$$eV = \frac{1}{2}m_e v^2$$

$$\Rightarrow v = \sqrt{\frac{2eV}{m_e}}$$

(b) From figure



$$R^2 + a^2 = (b - R)^2$$

$$\Rightarrow R^2 + a^2 = b^2 + R^2 - 2bR$$

$$\Rightarrow R = \frac{b^2 - a^2}{2b} \quad \text{... (i)}$$

Also, $R = \frac{m_e u_0}{eB_C}$ ... (ii)

From (i) and (ii)

$$\frac{b^2 - a^2}{2b} = \frac{m_e u_0}{eB_C} \Rightarrow B_C = \frac{2m_e b u_0}{e\left(b^2 - a^2\right)}$$

30. **Ans.** dBQ_0

Sol. $IdB = m\frac{dv}{dt}$

$dB(Idt) = mdv$

$$\Rightarrow dB\int_0^{Q_0} Idt = m\int_0^{v} dv$$

$$\Rightarrow dBQ_0 = mv$$

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